University of Pisa

Department of Physics "Enrico Fermi"

Global transport in soaring flights

Measurements, controls and questions for the supervisors

Matteo Di Sante • 12 September 2026

Chapter 3 • Complete chapter review

This deck develops the whole chapter without a prescribed duration. Its central distinction is between measured constraints on the observed ground trajectories and hypotheses about their generating process.

$$\mathbf{X}_f(t,\tau) = \mathbf{r}_f(t+\tau) - \mathbf{r}_f(t), \qquad R_f = |\mathbf{X}_f|.$$

$$M_2^{\mathrm{flight}}(\tau) = \frac{1}{N_f(\tau)} \sum_f \underbrace{\frac{1}{n_f(\tau)} \sum_k R_{fk}^2}_{\text{mean within flight } f} .$$

Windows stay within a retained segment. The main comparison covers $10 \le \tau \le 10^4$ s.

Retained and eligible populations

Archive: 156,406 paragliders and 6,094 hang gliders.

Full diagnostics: 155,085 and 6,060 eligible flights, all supported segments, on a 10-s grid.

Displacement growth, distribution shape, directional preference and memory must be confronted together.

Archive TAMSD: equal segment weights. Full V_1, V_2: equal flight weights. Baseline quantiles and moments: pooled windows.

What motion should a stochastic model reproduce?

The full diagnostic collector has no flight cap and no longest-segment-only restriction. Flights recorded more slowly than ten seconds are ineligible for its common grid. Each lag uses every supported flight; a separate fixed long-flight comparison controls population change. Different grids, minimum segment sizes and weights explain some differences from the archive TAMSD.

Launch, segment and flight averages use different weights

Estimator	Unit receiving equal weight	Support requirement
Launch ensemble	available flights at elapsed time t	observed coordinate at that time
Archive TAMSD	eligible retained segments	native-grid displacement within segment
Full V_1, V_2	eligible flights at lag τ	within-segment common-grid stencil
Pooled moments	available increment origins	stated non-overlapping baseline stencil

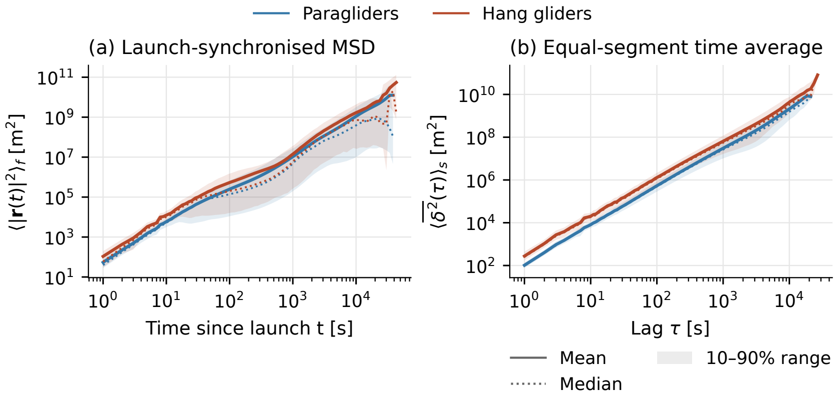
Different exponents can reflect different weights and changing support. Their difference alone is not an ergodicity test.

Which estimator corresponds to the physical population the model should reproduce?

Launch, segment and flight averages use different weights

Within a flight, contributions are combined using their admissible origins. The archive TAMSD and the full diagnostic reporter use different grids and minimum segment sizes, so their flight counts need not be identical. State the estimator before comparing numerical slopes.

Launch averages and time averages use different populations

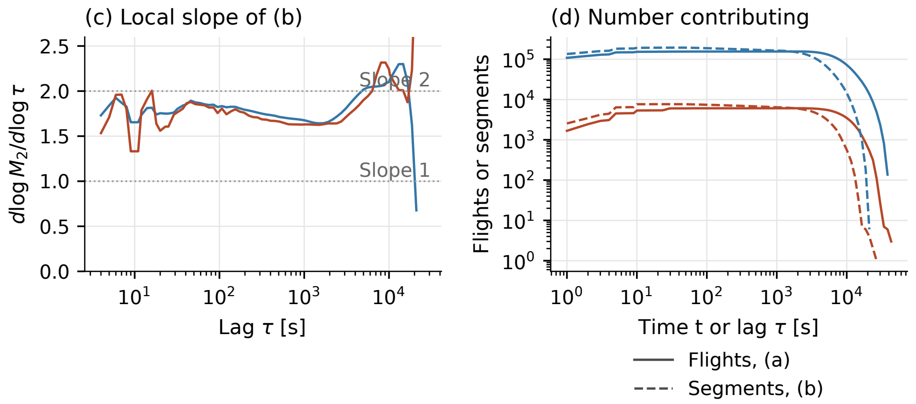


How much of the difference follows launch synchronization and segment weighting?

Launch averages and time averages use different populations

These are the upper two panels of the complete archive MSD figure. Blue denotes paragliders and terracotta hang gliders. The quantities use different clocks and weighting; their comparison is not an ergodicity test.

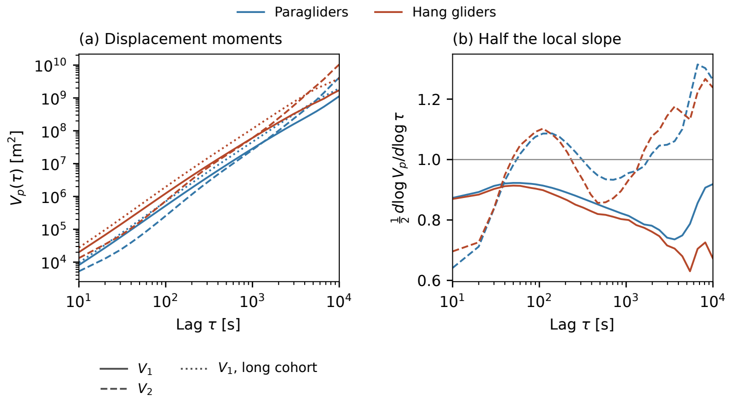
Local slopes must be read together with falling support



Which scale interval supports a stable finite-range description?

Local slopes must be read together with falling support

The lower panels of the archive MSD figure show local slopes and contributing flights or segments. A small bootstrap error does not remove scale dependence or population selection.



	b_1	b_2
Para	1.703	2.003
Hang	1.649	1.985

$V_1 = \langle R^2 \rangle$; V_2 uses second differences.

The local slope bends. One fitted number does not establish a power law.

Full eligible archive; 10–10,000 s. Solid: V_1; dashed: V_2; dotted: V_1, fixed long cohort. Right: half the local slope. Descriptive fits; no calibrated confidence intervals.

The MSD grows faster than linearly on the measured interval

The displayed exponents are imported from the completed full report. The first- and second-difference estimators use equal flight weights and do not cross segment boundaries. Their local slopes bend, so these are descriptive fits over the stated range. They differ from the equal-segment archive TAMSD and from pooled-origin high-order moments.

Constant velocity cancels; an intrinsic H remains open

$$A_2\mathbf{r} = \mathbf{r}(t+2\tau) - 2\mathbf{r}(t+\tau) + \mathbf{r}(t), \qquad V_2 = \langle |A_2\mathbf{r}|^2 \rangle.$$

Exact cancellation

$$\mathbf{r}(t) = \mathbf{r}_0 + \mathbf{v}_d t + \mathbf{Y}(t) \quad \Rightarrow \quad A_2\mathbf{r} = A_2\mathbf{Y}.$$

The constant velocity may combine wind, route choice and net progress. A turn or spatially varying wind survives.

Conditional interpretation

If the centred residual has stationary increments and variance $K\tau^{2H}$,

$$V_2 = K(4 - 2^{2H})\tau^{2H}, \quad 0 < H < 1.$$

At $H = 1$ the prefactor vanishes. Our $b_2 \simeq 2$ cannot be read as a validated $H \simeq 1$.

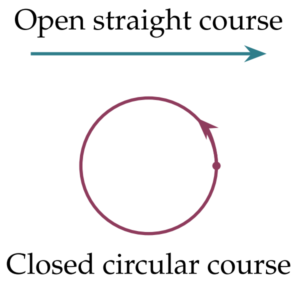
Which residual and which scale interval could satisfy the assumptions needed to estimate H?

At $\tau = 10^4$ s, V_2 retains only 14,360 para / 563 hang flights; it needs 2τ of continuous support.

Constant velocity cancels; an intrinsic H remains open

Explain the cancellation algebraically. Adjacent increments I1 and I2 obey Var(I1+I2)=K(2tau) to the power 2H. Expanding Var(I2-I1) gives the prefactor. The derivation requires finite variance and stationary increments with the stated power over the relevant lags, including 2tau. A smooth curved trajectory instead gives V2 proportional to tau to the fourth at small lag. We have therefore gained a diagnostic invariant, but have not isolated the motion in still air or obtained its Hurst exponent.

Open and closed tasks change the growth of displacements



Full eligible archive

	Task	N	b_1	b_2
Para	Open	67003	1.770	1.937
	Closed	87807	1.623	2.036
Hang	Open	2183	1.688	1.892
	Closed	3877	1.628	2.007

The ordering reverses at order two. Constant-drift cancellation leaves the return geometry in the observable.

Compare tasks at matched duration and region, using both physical lag and τ/T.

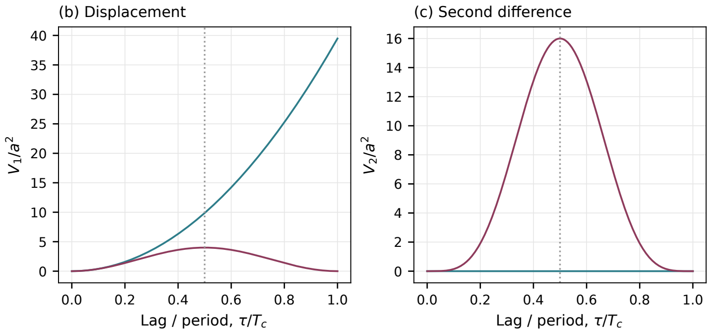
Task declarations; descriptive slopes, 10–10,000 s. Drawing is a geometric illustration. Exact circular example.

Open and closed tasks change the growth of displacements

The task declaration supplies open/closed labels; it does not force every retained segment to return to its start. Counts and slopes are read from the full-run report. Match duration and environment before interpreting task contrasts. The exact circular example illustrates how geometry survives second differencing; it is not a fitted flight model.

For $\mathbf{r}(t) = a(\cos \omega t, \sin \omega t)$ and $T_c = 2\pi/\omega$,

$$V_1 = 4a^2 \sin^2(\pi\tau/T_c), \qquad V_2 = 16a^2 \sin^4(\pi\tau/T_c).$$



Which scale effects could follow return geometry without a change of stochastic law?

A circle retains its geometry after second differencing

This is an exact periodically continued circle, not a fitted flight model. Constant speed does not imply ballistic endpoint displacement over every lag. The original two variation panels are retained; the circular path was illustrated on the preceding task slide.

Observed-path closure needs special care for segmented flights

$$c_f^{\text{obs}} = \frac{\left|\mathbf{r}_{f,\text{last}} - \mathbf{r}_{f,\text{first}}\right|}{\sum_{s\in\mathcal{S}_f}\sum_{k}\left|\mathbf{r}_{fs,k+1} - \mathbf{r}_{fs,k}\right|}.$$

The denominator omits unobserved intervals. For a segmented flight, c_f^{obs} need not be bounded by one.

Declared task, observed endpoint separation and actual flown-route closure are related descriptions with different information requirements.

Does the task contrast persist when restricted to continuously observed flights?

Observed-path closure needs special care for segmented flights

The descriptor preserves recorded endpoints without inventing missing path length. Reacquisition offsets can affect it even though every increment used for transport remains inside one segment. Use the task labels and this observed-path ratio as distinct descriptive stratifications; neither is an independent route-quality label.

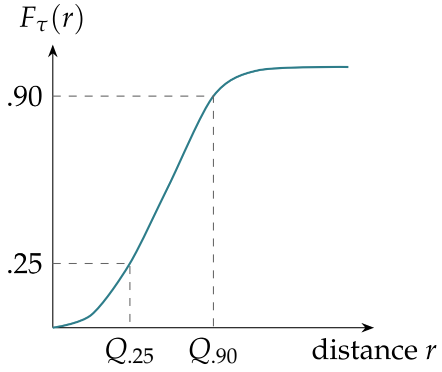
A single displacement scale predicts equal quantile slopes

$$Q_p(\tau) = \inf\{r : F_\tau(r) \geq p\}.$$

An H-self-similar process with stationary increments predicts

$$R(\tau) \stackrel{d}{=} (\tau/\tau_0)^H R(\tau_0),$$
$$Q_p(\tau) = (\tau/\tau_0)^H Q_p(\tau_0).$$

$H_p = H$ for every p; $\quad Q_{.90}/Q_{.25}$ stays fixed.



Schematic CDF. Quantiles describe ranks within one distribution; they do not define permanent classes of slow and fast flights. Equal slopes are necessary, not sufficient, for process self-similarity.

A single displacement scale predicts equal quantile slopes

The lower quartile probes relatively small displacements, while the 90th percentile probes larger ones. Both use the same distribution at each lag. A different flight may supply an observation near each rank, and rank identities may change; that does not invalidate a quantile. The mathematical issue is whether the population and weighting used to construct the distribution change with lag. No isotropy assumption is needed for the radial scaling implication.

The full archive and the fixed cohort answer different questions

Population	Paragliders	Hang gliders
Retained cleaned flights	156,406	6,094
Eligible for the common-grid diagnostic	155,085	6,060
Fixed long-flight control	14,360	563

Every eligible segment contributes where it supports the lag. The long-flight control adds a common population and common origins across the complete lag range.

Which statement concerns the observed archive, and which is conditional on long flights?

The full archive and the fixed cohort answer different questions

The common ten-second grid requires native cadence at most ten seconds and sufficient within-segment support. These are eligibility conditions, not flight-number caps imposed by the Mac. The twenty-thousand-second criterion is an operational choice for the additional control. It is not mathematically necessary merely to observe one increment at a ten-thousand-second lag.

The fixed control requires common temporal support

All eligible flights contribute at each supported lag in the archive-wide analysis. The additional fixed cohort contains every flight with a retained segment lasting at least 20,000 s.

This provides at least 1001 common ten-second origins in one qualifying segment at the largest lag of 10,000 s. Shorter segments of those flights contribute wherever they support the same full lag stencil.

How different are the shorter-flight results on a shorter common lag interval?

The fixed control requires common temporal support

The 20,000-second threshold is more restrictive than mere existence of a 10,000-second increment. It is an explicit support convention for this sensitivity analysis. The need for long observations is mathematical; the particular factor of two and chosen maximum lag are design choices, not necessities imposed by memory limits.

Holding flights fixed does not by itself hold the observations fixed

1. All available flights and origins, pooled.
2. Fixed flights, still pooling their available origins.
3. Fixed flights with equal total flight weight.
4. Fixed flights, equal flight weight and the same origins at every lag.

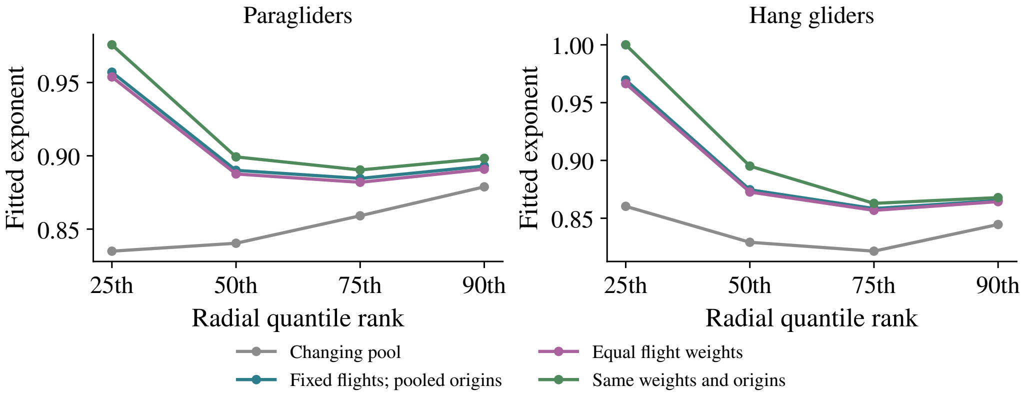
$$F_\tau(r) = \frac{1}{N_*} \sum_f \frac{1}{n_f^*} \sum_{k \in \mathcal{O}_f^*} \mathbf{1}\{R_{fk}(\tau) \le r\}.$$

How much of a change in the quantile curve comes from membership, weights or origin selection?

Holding flights fixed does not by itself hold the observations fixed

All common origins stay within a retained segment for every requested lag. Overlapping increments are deliberately retained; they are not independent replicates. The final control uses 14360 paraglider flights with 21739115 common origins and 563 hang-glider flights. Read the final hang-glider origin count from its completed report rather than a preflight estimate.

Membership, flight weights and common origins change the comparison

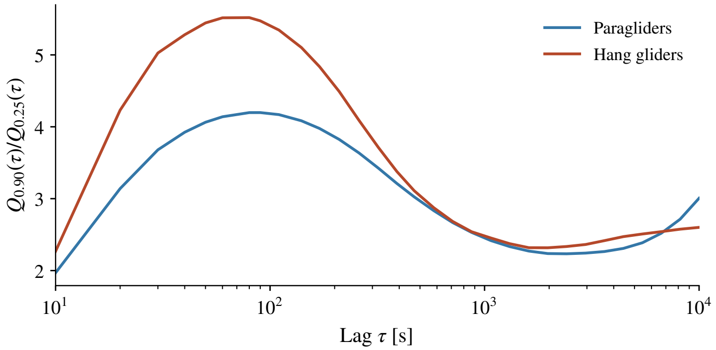


Which part of the slope change comes from each population control?

Membership, flight weights and common origins change the comparison

The four controls use the same ranks and fitting interval. The fixed cohort includes every qualifying long flight. Curves here are descriptive point estimates; the final fixed-control uncertainty uses the whole-flight paired bootstrap.

Relative spread changes, but the change is not monotone



$$\frac{d\log(Q_{.90}/Q_{.25})}{d\log\tau} = H_{.90}(\tau) - H_{.25}(\tau).$$

The ratio first rises, then falls, then rises again.

A negative fitted slope summarizes part of that evolution; it does not imply narrowing at every lag.

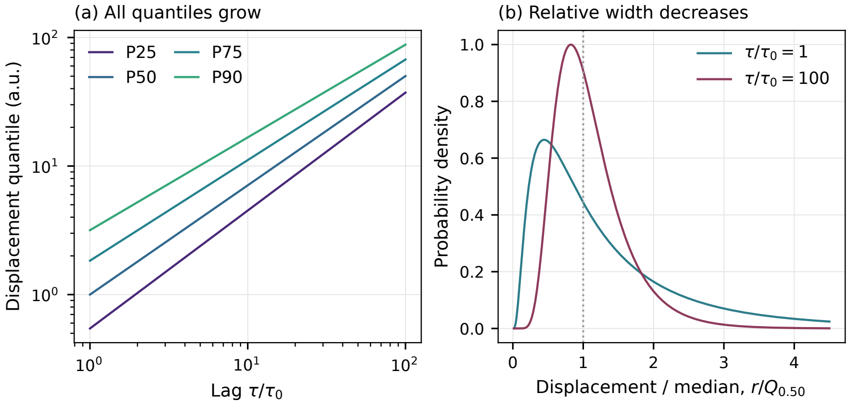
How much of this shape evolution survives site–day uncertainty, route matching and phase conditioning?

Fully controlled population. A lower ratio means less relative spread by this measure; it does not establish a flatter density tail.

Relative spread changes, but the change is not monotone

The fixed-control ratio starts near 1.97 for paragliders and 2.27 for hang gliders, peaks near ninety and eighty seconds, and has a post-peak minimum near two thousand seconds. The final values near 3.01 and 2.60 exceed the initial ones. A negative fitted log-ratio slope is therefore not monotone narrowing.

All quantiles can grow while spread relative to the median narrows

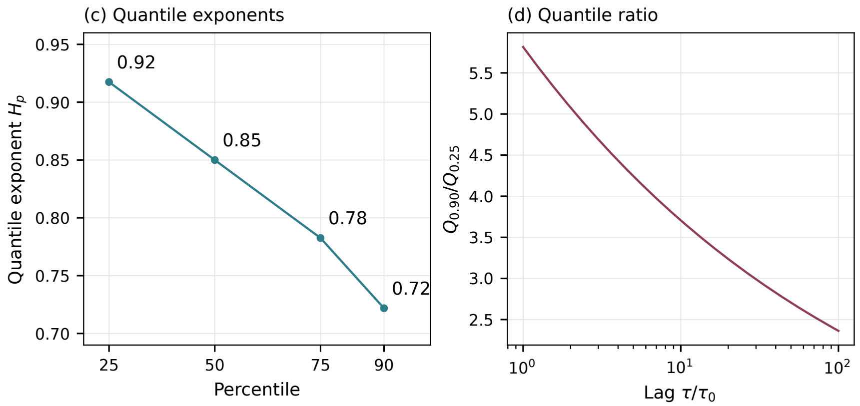


How can a changing distribution shape coexist with increasing displacement scales?

All quantiles can grow while spread relative to the median narrows

Analytical illustration: $R(\tau) = (\tau/\tau_0)^{0.85} \exp[\sigma(\tau)Z]$, with standard Gaussian Z and $\sigma(\tau) = 0.9 - 0.1 \log(\tau/\tau_0)$ on the illustrated range. This is not a fit to flight data.

Unequal rank exponents can encode that changing shape



Which empirical observations would distinguish this example from a true common scale family?

Unequal rank exponents can encode that changing shape

For the same lognormal illustration, Hp=0.85-0.1 Phi-inverse(p). The example explains the algebra without assigning permanent slow/fast identities to particular flights. The actual empirical curves need their own population controls and uncertainty estimates.

Anisotropy does not prevent a common scalar rescaling

$$\mathbf{X}(\tau) \stackrel{d}{=} \tau^H \mathbf{Z} \quad \Longrightarrow \quad |\mathbf{X}(\tau)| \stackrel{d}{=} \tau^H |\mathbf{Z}|,$$

even if the law of $\mathbf{Z}$ is anisotropic.

With two centred independent Gaussian components instead,

$$\mathbb{E}[R^2] = a\tau^{2H_E} + b\tau^{2H_N},$$

unequal component exponents can produce a lag-dependent radial effective exponent.

Is a mismatch caused by unequal directional scales, changing marginal shape or changing joint structure?

Anisotropy does not prevent a common scalar rescaling

Scalar self-similarity and isotropy are separate hypotheses. Operator self-similarity allows a matrix exponent and is broader, but does not permit arbitrary lag-by-lag normalization chosen to force collapse. The current plots assess the stated scalar comparison; they do not exhaust that wider model class.

East, north and radius together still do not identify the joint law

$$\{(1,2),(-1,-2)\} \qquad \text{versus} \qquad \{(1,-2),(-1,2)\},$$

each pair equiprobable.

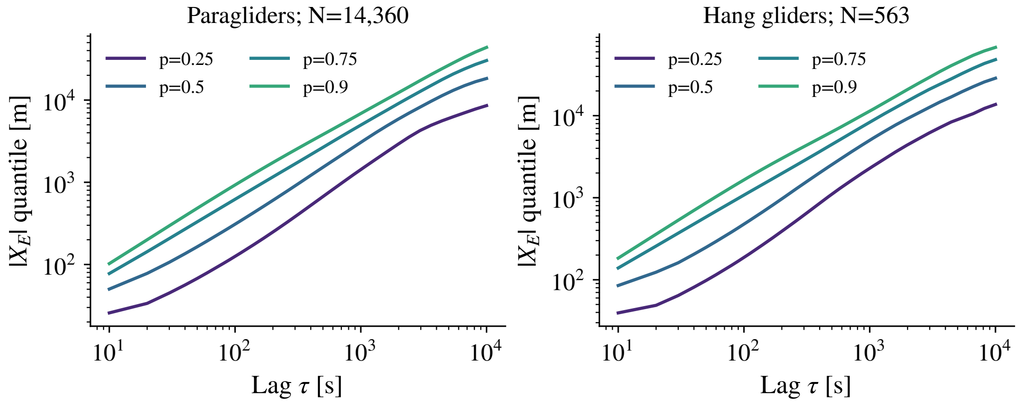
Quantity	First law	Second law
Signed east marginal	± 1	± 1
Signed north marginal	± 2	± 2
Radius	$\sqrt{5}$	$\sqrt{5}$
Cross-covariance	$+2$	-2

The missing information is real: signs and component dependence require a joint diagnostic.

East, north and radius together still do not identify the joint law

This is an exact mathematical counterexample, not a simulated flight result. Even signed marginals and radius can agree while the joint laws differ. The Cramer–Wold characterization uses all signed linear projections, not only the two coordinate projections. The signed joint plots address additional structure at their stated bin resolution.

Fixed-population quantiles: east component

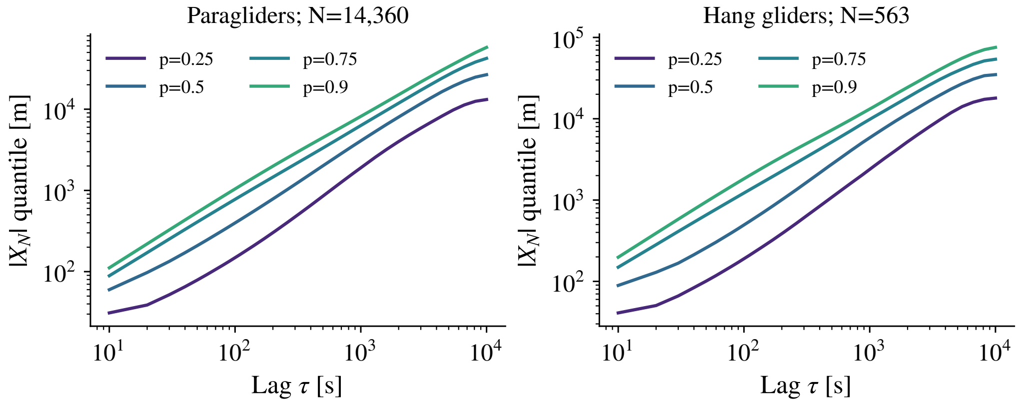


Do the ranks retain one relative shape as the lag increases?

Fixed-population quantiles: east component

Both disciplines use the same four ranks and their own complete fixed cohort. East and north use absolute signed-component magnitudes; radius is endpoint separation, not flown path length. Flights, origins and weights remain fixed across lags.

Fixed-population quantiles: north component



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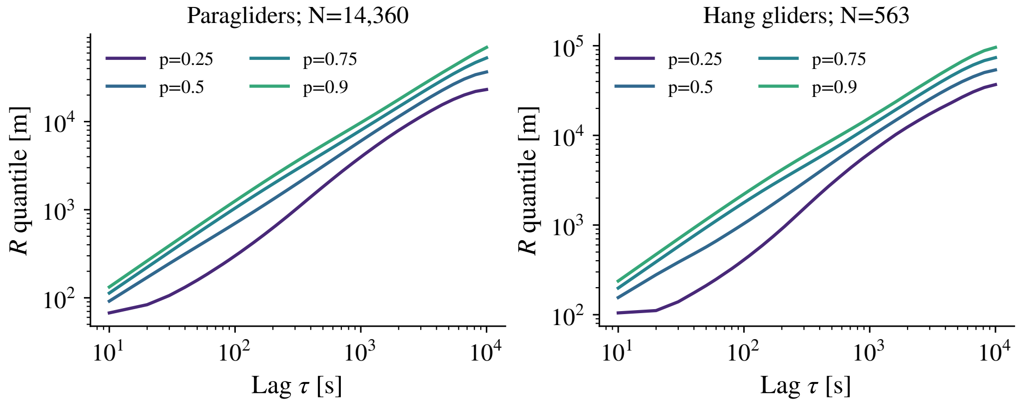
Fixed-population quantiles: radial displacement



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Fixed-population quantiles: radial displacement

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Whole-flight bootstrap preserves pairing across ranks, axes and lags

Resample flights with replacement 400 times. Each selected flight retains all its common origins and its joint east–north observations at every lag.

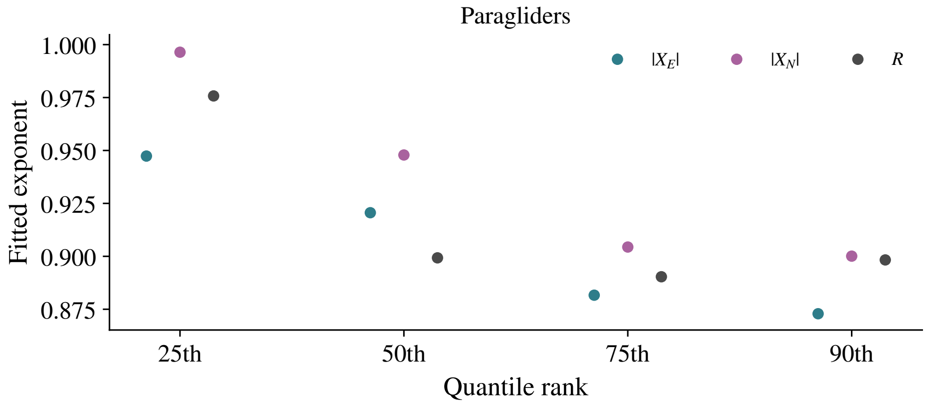
Overlapping origins remain within the resampled flight; they are not treated as independent replicates.

How would shared pilot history or site–day weather change these uncertainty estimates?

Whole-flight bootstrap preserves pairing across ranks, axes and lags

The current intervals assume independent flights within the fixed long-flight population. They are not iid increment intervals, simultaneous bands over all comparisons, or measurement-error intervals. Paired differences are computed inside each replicate. A large complete archive does not remove the assumptions behind the resampling model.

Paragliders: component and radial exponent estimates

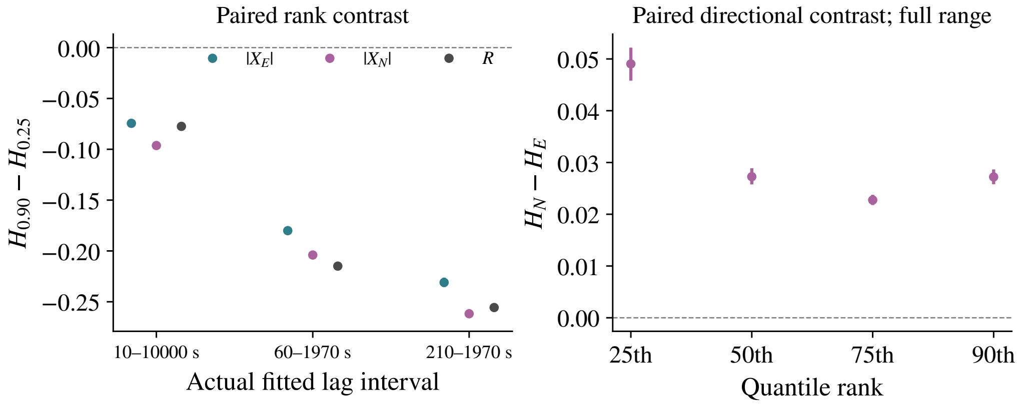


Which differences need a paired contrast instead of comparing overlapping intervals?

Paragliders: component and radial exponent estimates

Markers are full-range fitted exponents; vertical segments are the exact saved 95-percent percentile endpoints from 400 whole-flight bootstrap draws. They are not simultaneous intervals and do not include sensor, site-day or fitting-range uncertainty.

Paragliders: paired rank and directional contrasts

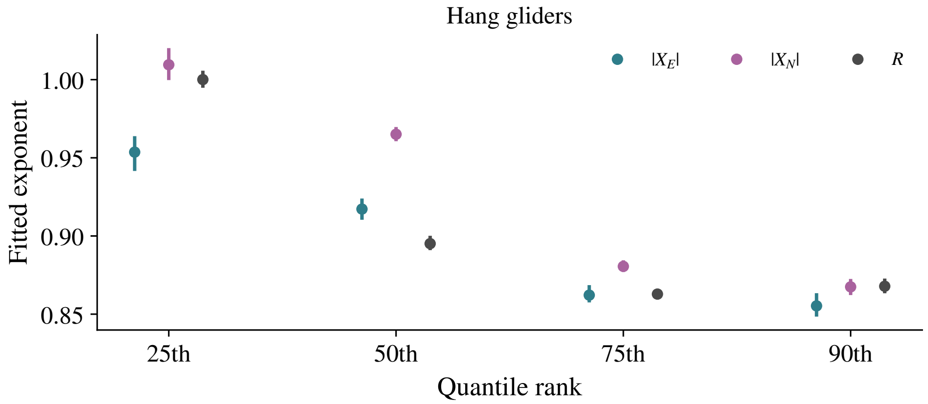


Which differences persist under a change of fitting interval?

Paragliders: paired rank and directional contrasts

The full-range north-minus-east contrast is positive with a 95-percent paired interval above zero at every one of the four ranks, in both disciplines. The rank contrast is H90 minus H25 for east, north and radius over each saved fit range. The directional contrast is north minus east at each rank over the full range. Both retain pairing inside the flight bootstrap. The zero line marks the stated equality hypothesis, without a multiple-comparison calibration.

Hang gliders: component and radial exponent estimates



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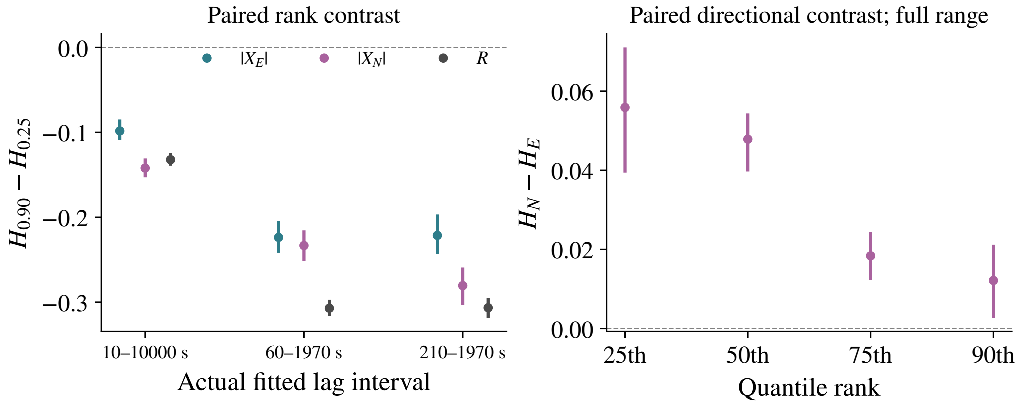
Hang gliders: paired rank and directional contrasts



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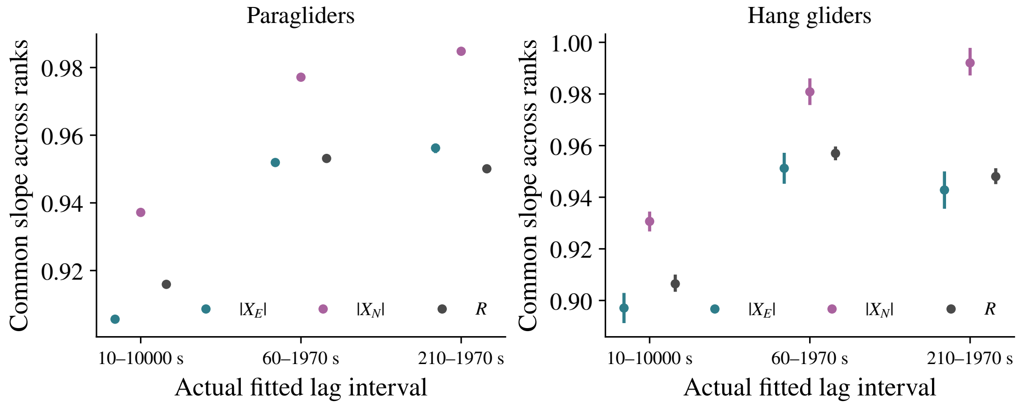
A common fitted scale depends on the fitted interval



Which range has an independent physical justification?

A common fitted scale depends on the fitted interval

The component-specific common slopes average the four ranks with their own intercepts. The plot uses saved fits and intervals only, not new fitting. Agreement of a common summary does not establish equal individual rank slopes.

$$\widehat{f}_j = \frac{p_j}{b_{j+1} - b_j}.$$

A very narrow histogram bin can have a tall density and still contain little probability. Across the displayed marginal laws, the mass below 1 mm, including zero and the straddling bin, is below 0.003%.

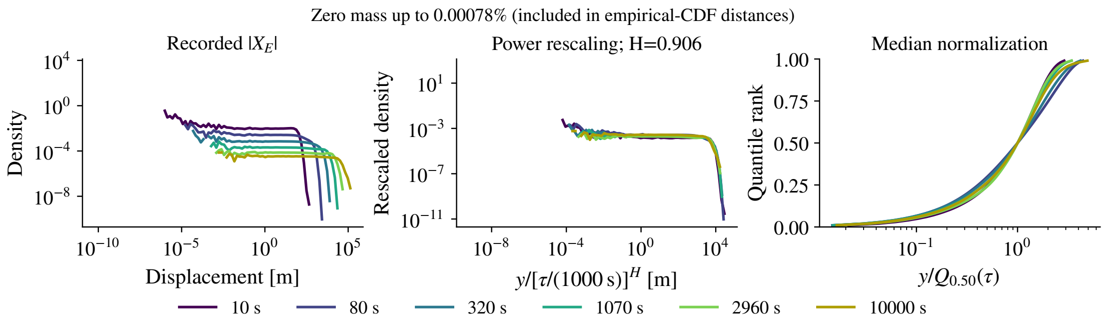
That numerical scale is not a claim of millimetre GNSS resolution. The following plots retain positive support; empirical-CDF comparisons also retain zero mass.

Do the central ranks and the probability-bearing tails tell the same scaling story?

A large density near zero need not carry much probability

The stated bound comes from the complete saved histogram masses, not from reading a logarithmic density height. Tiny numerical displacements can reflect recorded coordinates and interpolation. The median-normalized displays show the saved first-to-ninety-ninth percentiles; they do not establish tail collapse beyond those ranks.

Paragliders: marginal collapse for east

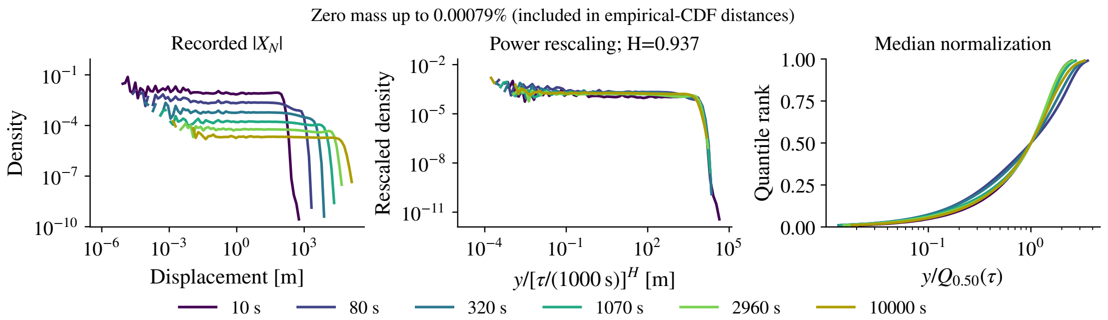


Does power or median normalization leave a systematic shape change?

Paragliders: marginal collapse for east

The panels show recorded histogram densities, the saved power rescaling, and recorded first-to-ninety-ninth quantile ranks after median normalization. Positive values are displayed on logarithmic axes; any zero mass is reported separately and retained in the exact empirical-CDF distances. Median-normalized ranks do not establish collapse outside that rank range. The same selected lag colours are retained throughout.

Paragliders: marginal collapse for north



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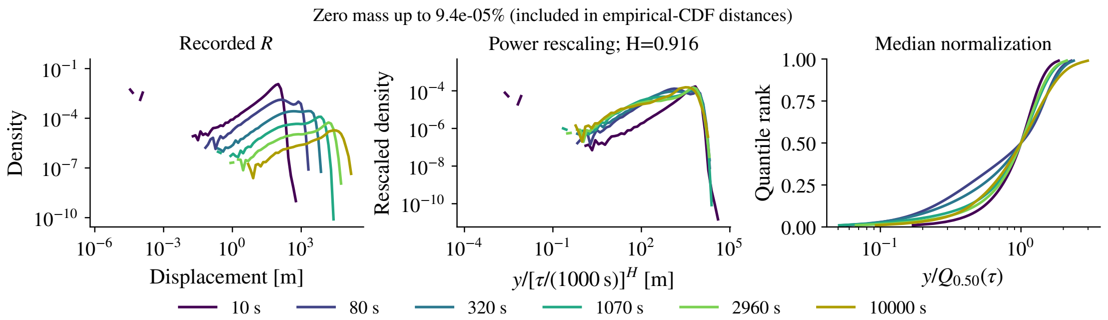
Paragliders: marginal collapse for radius



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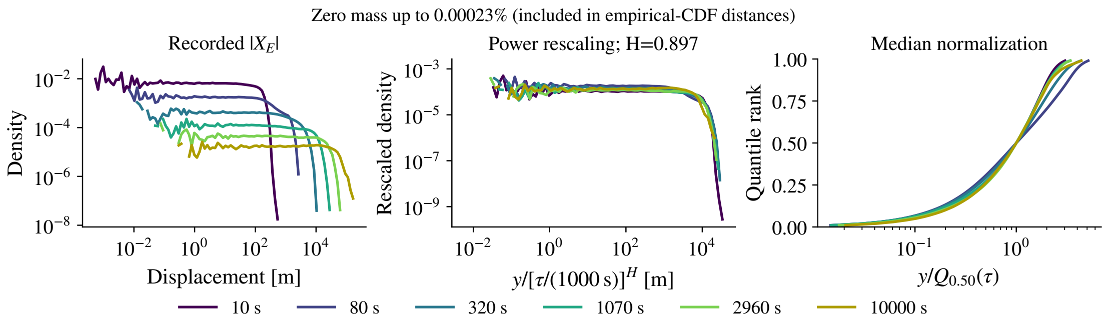

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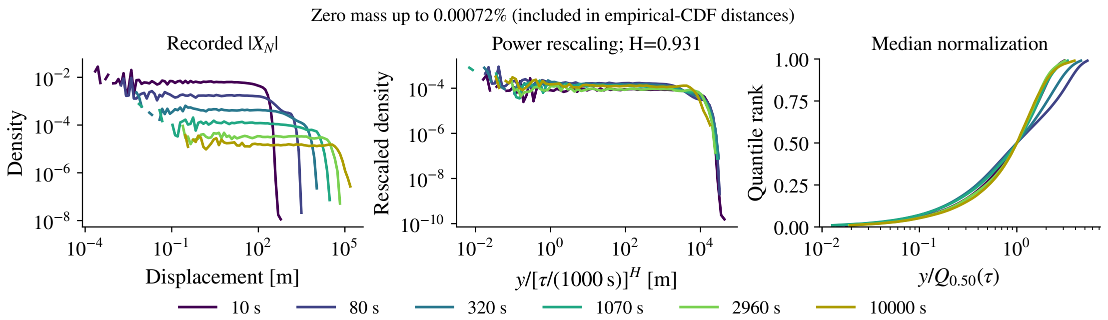

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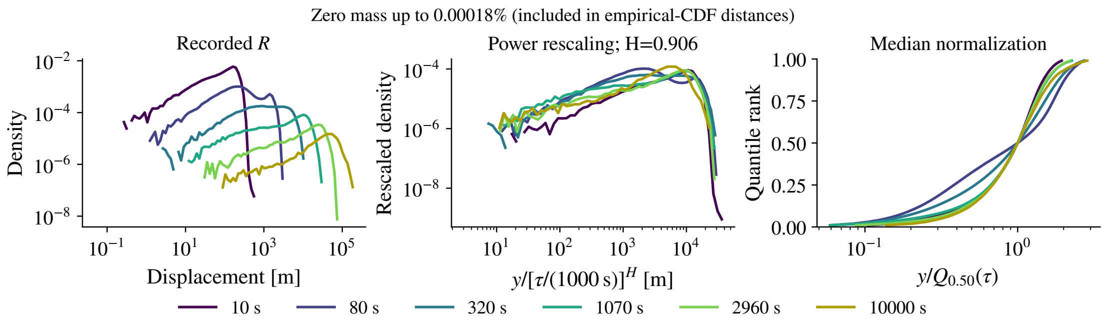

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Hang gliders: marginal collapse for radius

The panels show recorded histogram densities, the saved power rescaling, and recorded first-to-ninety-ninth quantile ranks after median normalization. Positive values are displayed on logarithmic axes; any zero mass is reported separately and retained in the exact empirical-CDF distances. Median-normalized ranks do not establish collapse outside that rank range. The same selected lag colours are retained throughout.

The joint comparison keeps geographic axes and one scalar scale

$$\mathbf{Z}_\tau = \frac{(X_E, X_N)}{s_0 \left(\frac{\tau}{1000\,\mathrm{s}}\right)^{H_*}}.$$

H_* is the common fit across all component/rank slopes. The radial reference median near 1000 s is adjusted to 1000 s by the same power.

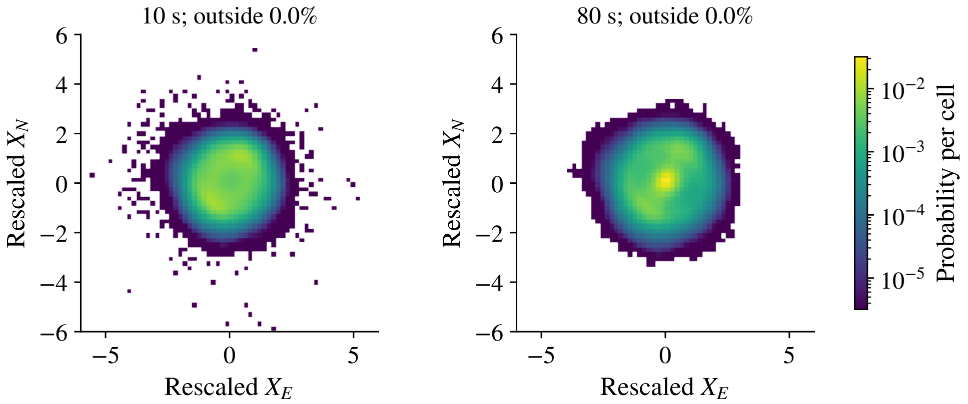
There is no lag-specific centring, rotation or whitening. The six lags share the same cells and equal total weight per flight.

Which directional and dependence changes remain visible when the normalization cannot remove them separately?

The joint comparison keeps geographic axes and one scalar scale

The common geographic axes are deliberate. Independent rotations or covariance normalization at each lag could erase part of the effect being studied. This is a test of one stated scale family of ground displacements, not a recovery of air-relative motion.

Paragliders: signed joint laws (1/3)

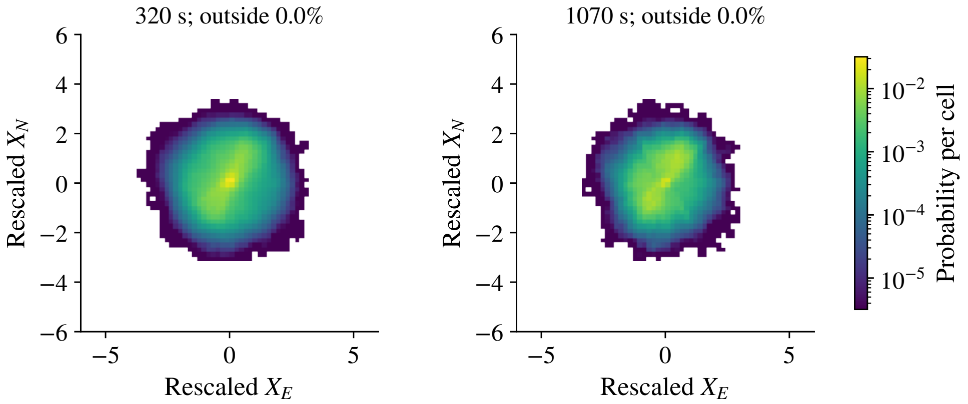


Which changes concern orientation, central concentration or mass outside the displayed square?

Paragliders: signed joint laws (1/3)

Each discipline uses one common colour scale across all six lags, one geographic frame and one scalar rescaling. Colours are probability per original cell. Outside mass is retained in overflow cells for the numerical comparison. Equal plot area does not imply equal precision across disciplines.

Paragliders: signed joint laws (2/3)

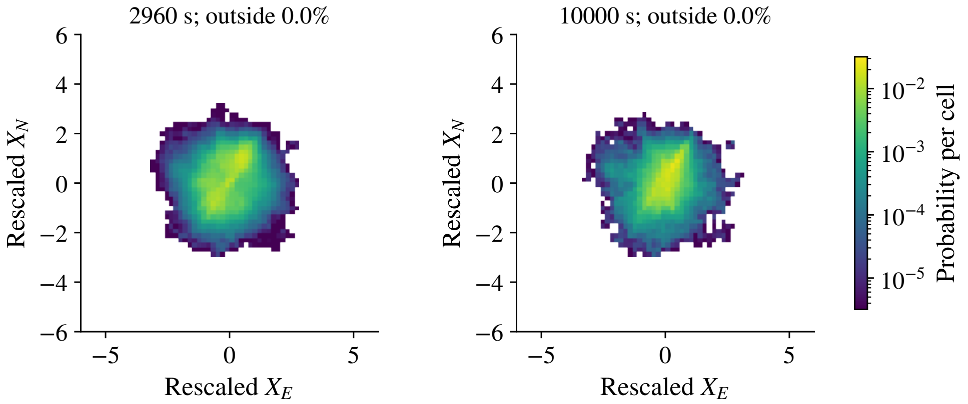


Which changes concern orientation, central concentration or mass outside the displayed square?

Paragliders: signed joint laws (2/3)

Each discipline uses one common colour scale across all six lags, one geographic frame and one scalar rescaling. Colours are probability per original cell. Outside mass is retained in overflow cells for the numerical comparison. Equal plot area does not imply equal precision across disciplines.

Paragliders: signed joint laws (3/3)

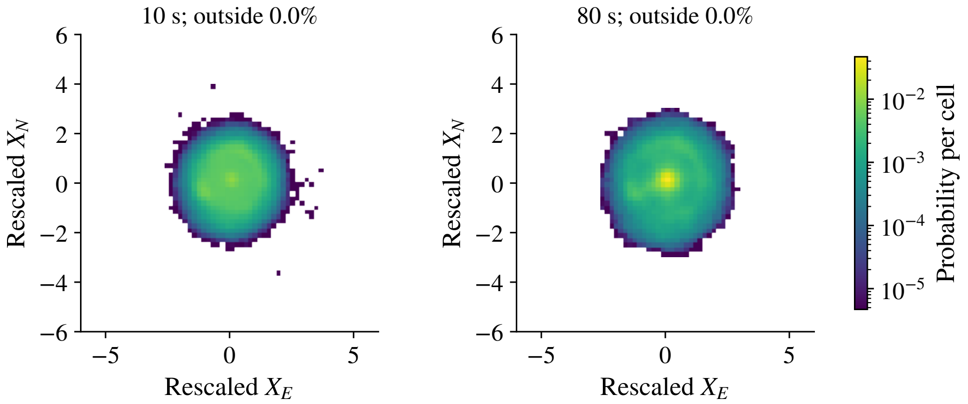


Which changes concern orientation, central concentration or mass outside the displayed square?

Paragliders: signed joint laws (3/3)

Each discipline uses one common colour scale across all six lags, one geographic frame and one scalar rescaling. Colours are probability per original cell. Outside mass is retained in overflow cells for the numerical comparison. Equal plot area does not imply equal precision across disciplines.

Hang gliders: signed joint laws (1/3)



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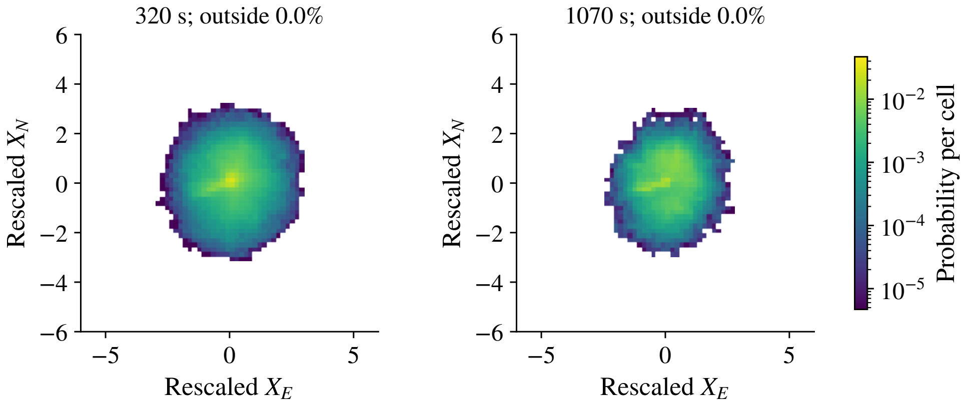
Hang gliders: signed joint laws (1/3)

Each discipline uses one common colour scale across all six lags, one geographic frame and one scalar rescaling. Colours are probability per original cell. Outside mass is retained in overflow cells for the numerical comparison. Equal plot area does not imply equal precision across disciplines.

Hang gliders: signed joint laws (2/3)



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Hang gliders: signed joint laws (2/3)

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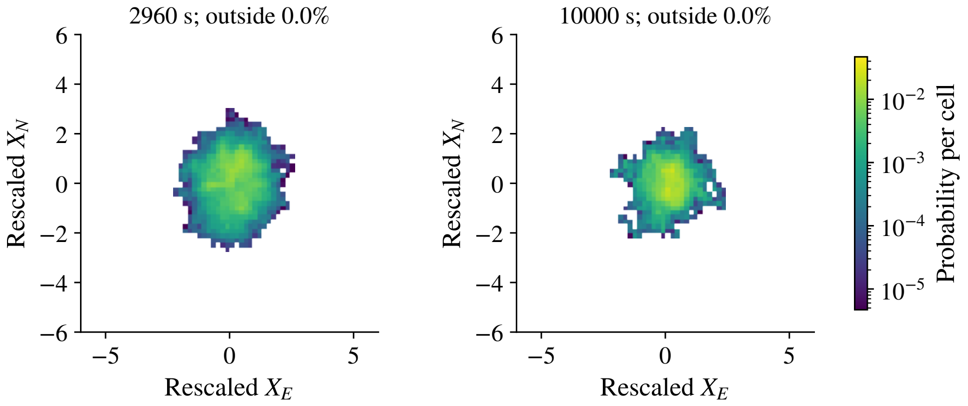

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Hang gliders: signed joint laws (3/3)

Each discipline uses one common colour scale across all six lags, one geographic frame and one scalar rescaling. Colours are probability per original cell. Outside mass is retained in overflow cells for the numerical comparison. Equal plot area does not imply equal precision across disciplines.

One scalar rescaling leaves changes in the signed joint law

	Paragliders	Hang gliders
Common scalar exponent H_*	0.9196	0.9114
Binned TV: 10 s versus 10,000 s	0.370	0.403
Largest TV among the six lags	0.370	0.441

Later rescaled laws concentrate more centrally, while their means also change. Overflow mass is zero to numerical precision in the displayed square.

Mean motion, centred spread, orientation and dependence can all contribute. The plot alone does not separate their physical causes.

Would separating mean displacement from centred shape clarify which assumption fails?

One scalar rescaling leaves changes in the signed joint law

The comparison uses fixed flights, common origins, equal flight weights, signed geographic components and the same bins. These distances describe the recorded finite distributions; they have no calibrated rejection threshold. The first-to-last pair is not the maximum pair for hang gliders.

One-lag vector laws leave multi-time questions open

$$D_{\text{joint}}(\tau, \tau') = \frac{1}{2} \sum_{ij} |p_{\tau,ij} - p_{\tau',ij}|.$$

The grid has 64 interior bins per axis and overflow cells. The distance resolves changes at that bin scale; it is not a calibrated rejection test.

Even agreement of every one-lag vector distribution would leave the joint law at several times, and hence full process self-similarity, untested.

Which multi-time or phase-conditioned comparison would most help choose between candidate processes?

One-lag vector laws leave multi-time questions open

Finite empirical distributions can differ under a common population law. Calibration must respect paired flights and dependent origins. Energy distance is an alternative multivariate diagnostic under finite first moments, but choosing another distance does not automatically supply the right uncertainty model.

Regional plots use different clocks, weights and centring

Diagnostic	Clock	Weighting	Centring
Launch-time ratio	time since launch t	paired flights at each t	raw moments
Regional PCA	increment lag τ	eligible origins pooled	mean removed
Fixed joint law	increment lag τ	equal total flight weight	mean retained

These measurements differ in method as well as population. Compare these choices before attributing a difference to a physical mechanism.

Which apparent disagreement remains after comparing like populations and observables?

Regional plots use different clocks, weights and centring

The PCA draws on the full eligible regional archive, not only the fixed long-flight cohort. Longer supported records supply more increment origins to that pooled covariance. The launch-time ratios use a different clock and a changing population. Make these distinctions before linking the three figures to wind or terrain.

Equal east and north moments do not establish isotropy

$$\rho_{EN} = 1, \qquad \Sigma = \begin{pmatrix} 1 & 0.9 \\ 0.9 & 1 \end{pmatrix}, \qquad \lambda_1/\lambda_2 = 19.$$

This centred Gaussian has balanced coordinate moments but strongly unequal principal variances.

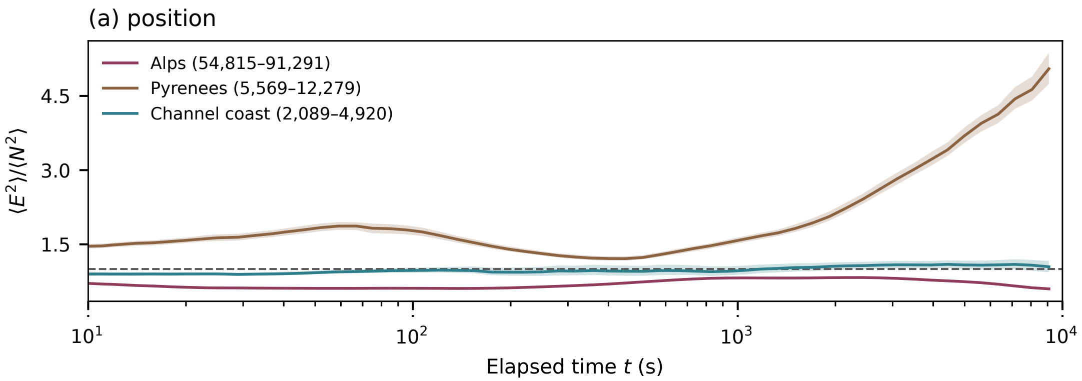
A pooled mixture of east-directed and north-directed routes can also balance the ensemble while each flight remains directional.

Are we observing mean drift, centred directional variability, or a mixture of individually oriented routes?

Equal east and north moments do not establish isotropy

Full isotropy requires rotation invariance of the law. An isotropic covariance alone would still be only a second-order condition. Raw moments additionally contain the outer product of the mean with itself. An intentional downwind route can be successful despite being directional, so this ratio is not a direct performance score.

Regional position moments differ along east and north

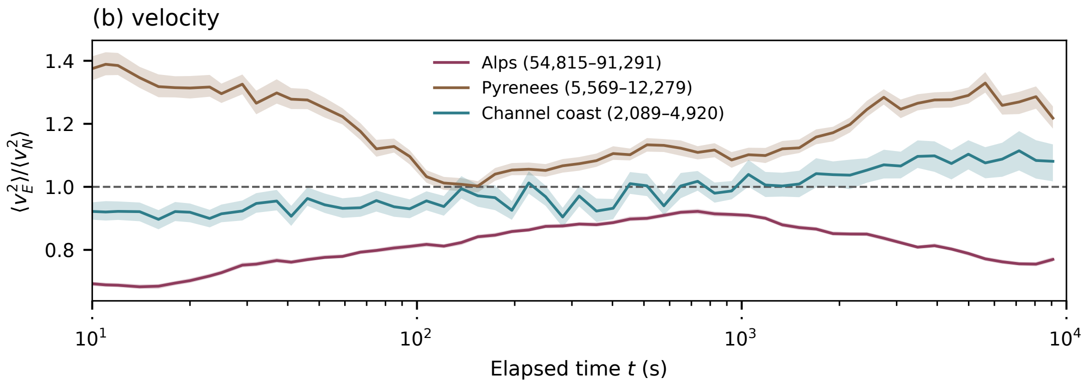


How much imbalance reflects mean drift, route orientation or changing support?

Regional position moments differ along east and north

The clock is time since estimated launch, not an increment lag. Position ratios approach five in the Pyrenees. Raw second moments include both covariance and the squared mean. Splits prevent cross-gap stencils but recorded endpoint offsets can still affect these launch-referenced coordinates.

Regional velocity moments show a related directional contrast



Which position and velocity contrasts persist under matched conditions?

Regional velocity moments show a related directional contrast

The regional figure pools paired east/north observations at each elapsed time and uses site-day pointwise bands. Terrain, weather, routes and the disappearing short-flight population can all contribute. Coordinate-moment balance is weaker than full isotropy.

Regional uncertainty bands do not match the groups

The ratios use paired east and north observations. A site–day bootstrap with 200 draws gives pointwise 10–90% ranges; each plotted time needs at least 30 flights and 10 clusters.

Missing site/date keys are treated as singletons. Resampling these groups does not match the equipment populations on weather, task or duration.

Would a paired site–day comparison change the group difference and its uncertainty?

Regional uncertainty bands do not match the groups

The bands are not 95-percent intervals, simultaneous bands or direct intervals for the difference between groups. Their overlap is not itself a calibrated significance test. The many displayed time points are correlated and must not be counted as independent confirmations of an ordering.

PCA describes preferred axes; wind needs independent information

A rotation preserves distance

$$\Sigma = U\Lambda U^{\top}, \qquad |U^{\top}\mathbf{X}| = |\mathbf{X}|.$$

PCA decorrelates components at a chosen lag. It cannot repair radial collapse or remove temporal dependence.

Whitening changes lengths and only fixes second moments at the fitted scale.

Ground velocity has two unknown terms

$$\mathbf{v}_{\text{ground}} = \mathbf{v}_{\text{air}} + \mathbf{w}_h.$$

Terrain constrains headings; wind also changes position. Differentiation does not separate the two.

Different marginal exponents, or changing joint dependence, can prevent radial scaling.

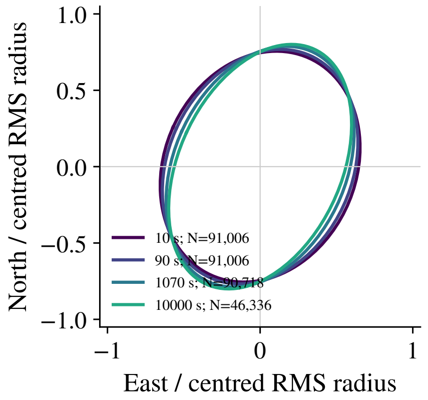
Choose matched environments and an independent wind estimate before defining an "environment-free" model target.

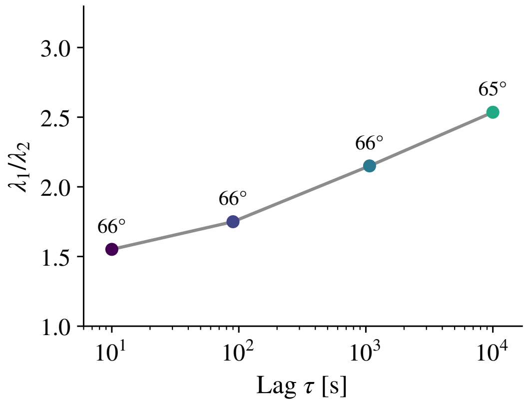
The following panels report regional PCA and its support. Anisotropy alone is compatible with a self-similar vector and a self-similar radius.

PCA describes preferred axes; wind needs independent information

Even independent Gaussian components can yield a changing radial exponent if their component exponents differ. If both component exponents agree, changing dependence can still prevent joint scaling. Conversely, a joint vector scaling law always implies radial scaling, whether the vector is isotropic or not. Rotating each flight randomly would balance aggregate directions without changing radial displacements; it cannot recover still-air motion. Independent wind information or carefully justified phase-based wind estimates should be discussed before choosing an environmental correction.

Alps: centred principal axes across four lags

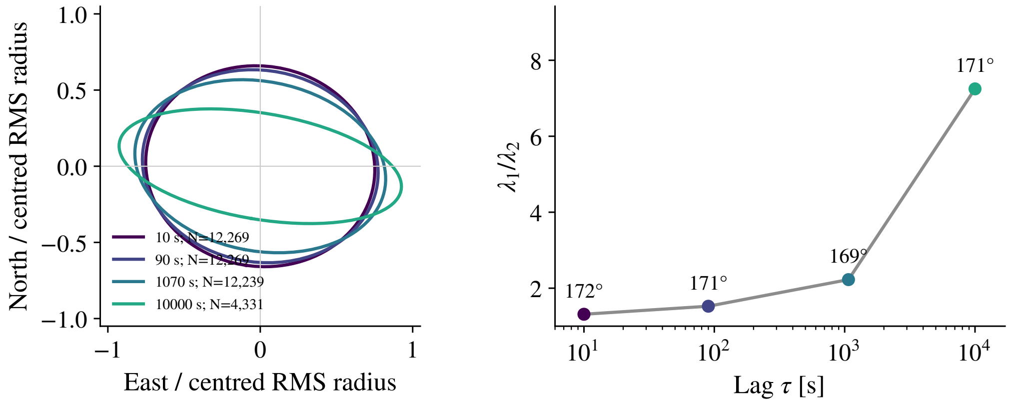




Does the axis remain stable as lag and contributing population change?

Alps: centred principal axes across four lags

The covariance pools eligible regional origins from the full archive. Ellipses are centred and normalized by the centred RMS radius; they hide absolute magnitude and drift. The axis is modulo 180 degrees and becomes poorly determined near equal eigenvalues. These panels display all four reported lags with their contributing-flight counts.

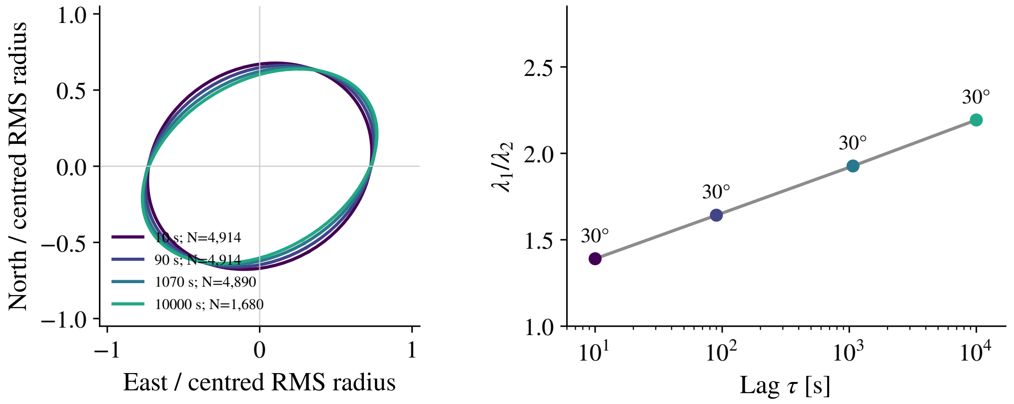


Does the axis remain stable as lag and contributing population change?

Pyrenees: centred principal axes across four lags

The covariance pools eligible regional origins from the full archive. Ellipses are centred and normalized by the centred RMS radius; they hide absolute magnitude and drift. The axis is modulo 180 degrees and becomes poorly determined near equal eigenvalues. These panels display all four reported lags with their contributing-flight counts.

Channel Coast: centred principal axes across four lags



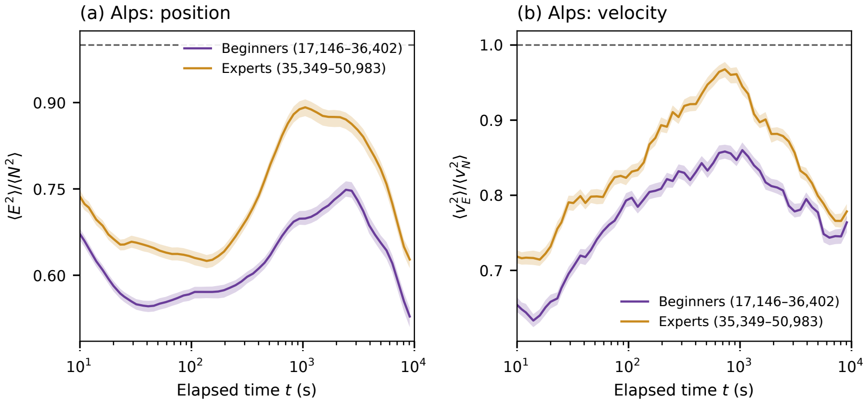
Does the axis remain stable as lag and contributing population change?

Chapter 3: global transport | Slide 54

Channel Coast: centred principal axes across four lags

The covariance pools eligible regional origins from the full archive. Ellipses are centred and normalized by the centred RMS radius; they hide absolute magnitude and drift. The axis is modulo 180 degrees and becomes poorly determined near equal eigenvalues. These panels display all four reported lags with their contributing-flight counts.

Alps: C/D/CCC has smaller coordinate-moment imbalance

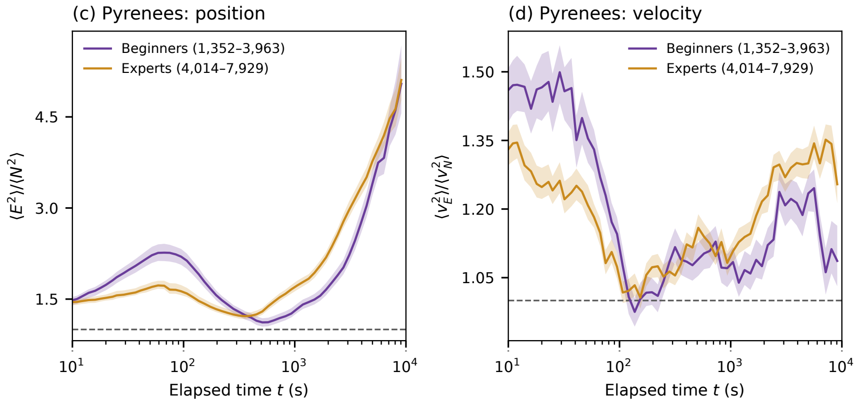


Does the ordering persist after matching site, day, task and duration?

Alps: C/D/CCC has smaller coordinate-moment imbalance

The fresh Alpine position and velocity ratios stay closer to unity for C/D/CCC over the supported grid. This is a descriptive ensemble observation. It does not establish individual isotropy, greater skill or a causal equipment effect. Violet: A/B; ochre: C/D/CCC. Beginners and experts in the retained figure legend denote equipment proxies. Bands are pointwise 10–90-percent site-day bootstrap ranges, not intervals for the group difference.

Pyrenees: the ordering of the two groups reverses

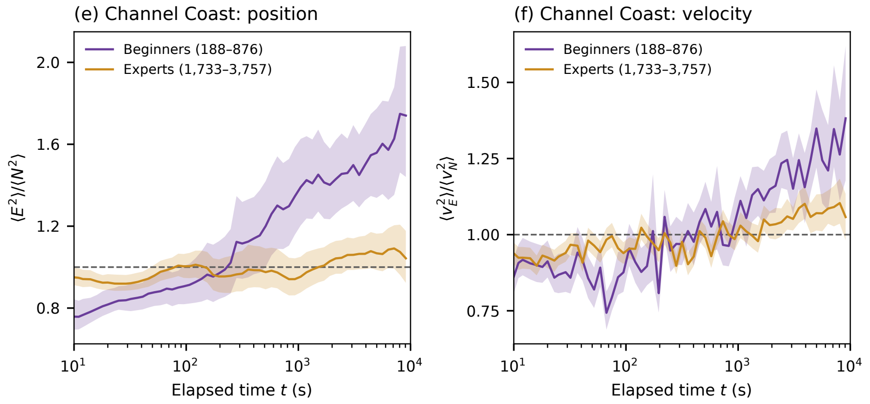


What changes between the early and later portions of the curves?

Pyrenees: the ordering of the two groups reverses

The Pyrenean curves do not support a uniform expert-proxy advantage. At late times both position ratios exceed five, while the velocity ordering also changes. Adjacent plotted times are correlated observations, not independent tests. Violet: A/B; ochre: C/D/CCC. Beginners and experts in the retained figure legend denote equipment proxies. Bands are pointwise 10–90-percent site-day bootstrap ranges, not intervals for the group difference.

Channel Coast: the late position contrast is pronounced

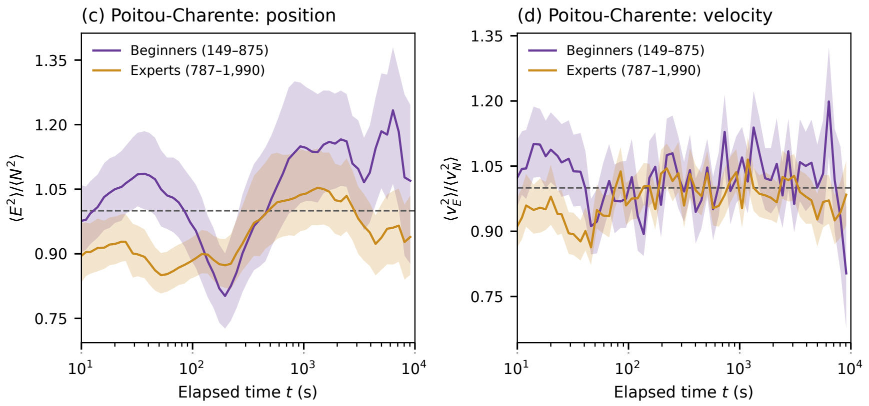


Can matching conditions separate directional route choice from wind response?

Channel Coast: the late position contrast is pronounced

Near 9086 seconds the position ratios are about 1.74 for A/B and 1.04 for C/D/CCC, supported by 188 and 1733 paired flights. These are raw ensemble second moments and the samples differ strongly. Coastal cliffs and valleys prevent a wind-only interpretation. Violet: A/B; ochre: C/D/CCC. Beginners and experts in the retained figure legend denote equipment proxies. Bands are pointwise 10–90-percent site-day bootstrap ranges, not intervals for the group difference.

Poitou-Charente: the group curves cross

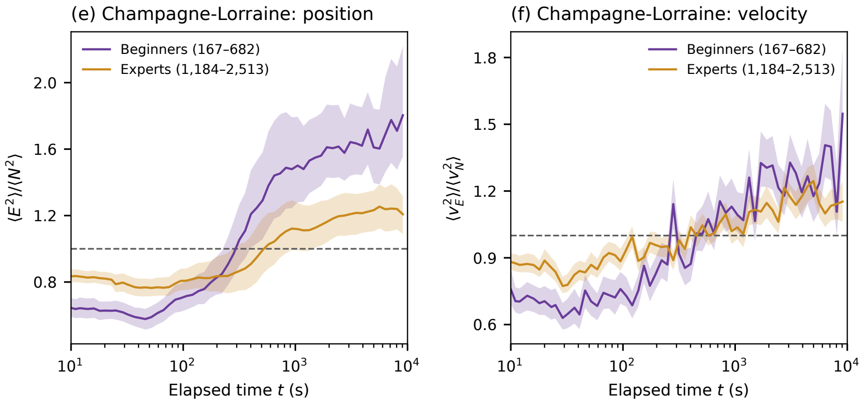


Does this low-relief launch box reproduce the coastal observation?

Poitou-Charente: the group curves cross

The crossing curves and broad overlapping bands do not support a general group ordering. These broad launch-location boxes are neither DEM-verified plains nor controls matched on weather. Violet: A/B; ochre: C/D/CCC. Beginners and experts in the retained figure legend denote equipment proxies. Bands are pointwise 10–90-percent site-day bootstrap ranges, not intervals for the group difference.

Champagne-Lorraine: a late contrast does not keep C/D/CCC at unity



Which parts of the coastal hypothesis transfer to this separate region?

Champagne-Lorraine: a late contrast does not keep C/D/CCC at unity

A later position contrast appears, but the C/D/CCC ratio also rises above one. Region, route mixture, weather and declining support must be separated before claiming the same physical cause. Violet: A/B; ochre: C/D/CCC. Beginners and experts in the retained figure legend denote equipment proxies. Bands are pointwise 10–90-percent site-day bootstrap ranges, not intervals for the group difference.

The environmental interpretation remains a hypothesis

Hypothesis to discuss
Equipment and pilot choices may moderate directional constraints associated with wind and terrain. Similar mountain profiles may reflect shared route constraints.

Evidence limiting that interpretation
The Pyrenean ordering reverses; class is an experience proxy; coastal terrain is not absent; late support differs; and ensemble balance does not establish individual isotropy or performance.

What observation would distinguish this hypothesis from different sites, tasks, weather or route mixtures?

The environmental interpretation remains a hypothesis

Preserve the proposed explanation as a useful hypothesis rather than silently removing it or presenting it as established. Similar curve shape does not by itself establish a common cause or an unavoidable terrain constraint. Wind and terrain can interact, so the coast/mountains split does not identify their separate effects.

Define the target before calling it resistance to wind

$$\mathbf{v}_{\text{ground}} = \mathbf{v}_{\text{air}} + \mathbf{w}_h.$$

Upwind progress, air-relative performance, choice of route and geographic moment balance are different outcomes.

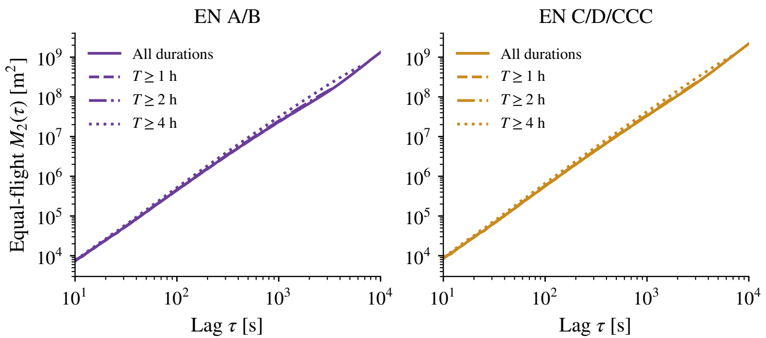
A first comparison would match overlapping site–day, duration and task support; then separate mean drift, centred variability and individual route orientation. Independent wind and local terrain axes would support environmental alignment.

Which target and matched comparison should be the first priority?

Define the target before calling it resistance to wind

The site–day bootstrap adjusts one uncertainty structure; it does not perform matching or remove confounding. Where data permit, the same pilot on different equipment could help assess selection, but weather and task still require control. Differentiating velocity removes a constant wind term, but not its variation along the trajectory or the pilot's manoeuvres.

Long flights occur in both equipment groups



For $T \geq 4$ h: 7,415 A/B flights and 27,815 C/D/CCC flights.
Across 10–10,000 s, all-duration slopes are 1.728 and 1.788.

Equal-flight MSD; groups are nested. Threshold curves stop at $\tau = T_0/2$; at least 30 flights per lag. All counts and common-range slopes.

Long flights occur in both equipment groups

The hypothesis that only experts contribute to flights longer than four hours is contradicted by the counts. Experts nevertheless become more frequent in the long-duration groups. On the common 10–1695 second interval, expert slopes exceed beginner slopes in every duration cell; within both classes slopes also increase between all flights and the four-hour group. This shorter interval is needed to compare all four nested cohorts under the half-duration rule. It is distinct from the main three-decade fit. Equipment, pilot selection, region and weather remain possible causes.

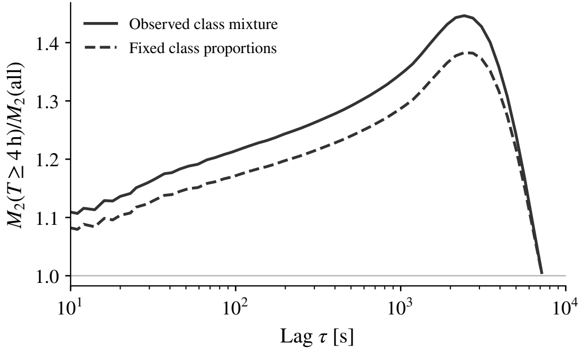
Duration and equipment on the common 10–1695 s interval

Experience proxy	Duration	Flights	b	RMS residual (dex)
Beginners	All	54,292	1.755	0.022
Beginners	$T \geq 1$ h	51,124	1.754	0.022
Beginners	$T \geq 2$ h	31,768	1.767	0.020
Beginners	$T \geq 4$ h	7,415	1.806	0.017
Experts	All	94,283	1.803	0.017
Experts	$T \geq 1$ h	91,182	1.802	0.017
Experts	$T \geq 2$ h	69,087	1.811	0.016
Experts	$T \geq 4$ h	27,815	1.836	0.015

Counts are eligible flights, not independent windows; support falls with lag. RMS residual measures departures from a fitted line, not uncertainty. Threshold samples overlap.

Duration and equipment on the common 10–1695 s interval

Fixing equipment proportions removes part of the duration contrast



$$M_{T_0}^{\mathrm{fixed}}(\tau) = \sum_c \pi_c M_{T_0,c}(\tau).$$

Fixed weights: $\pi_{\mathrm{A/B}} = 0.365$, $\pi_{\mathrm{C/D/CCC}} = 0.635$.

At $\tau = 1695$ s the ratio falls from 1.42 to 1.35. A within-class contrast remains.

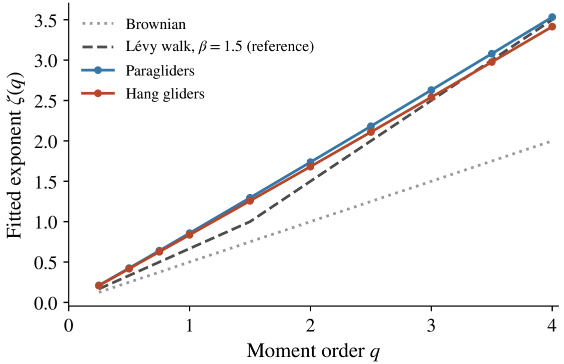
Match site–day and task next. Convergence near 7200 s also reflects shorter flights leaving the reference.

Both curves use the same two known equipment groups. Equality at one lag does not establish independence from flight duration.

Fixing equipment proportions removes part of the duration contrast

The expert fraction among known classes rises from 63.5 percent overall to 78.9 percent in the four-hour cohort. Standardization uses the all-duration proportions at every lag for both numerator and denominator. It is a descriptive reweighting, not a causal adjustment. The remaining difference shows that the two-class proportions alone cannot account for the observed amplitude contrast. Near half the duration threshold, the reference loses shorter flights, so the curves are increasingly comparing similar populations.

The moment spectrum challenges a specific Lévy-walk benchmark



$M_q = \langle R^q \rangle \propto \tau^{\zeta(q)}$.

For a constant-speed renewal walk with isotropic independent headings and $\psi(t) \sim t^{-1-\beta}$, $1 < \beta < 2$:

$$\zeta_{\mathrm{LW}}(q) = \begin{cases} q/\beta, & q < \beta, \\ q + 1 - \beta, & q > \beta. \end{cases}$$

The empirical spectrum is nearly linear over $q = 0.25$–4.

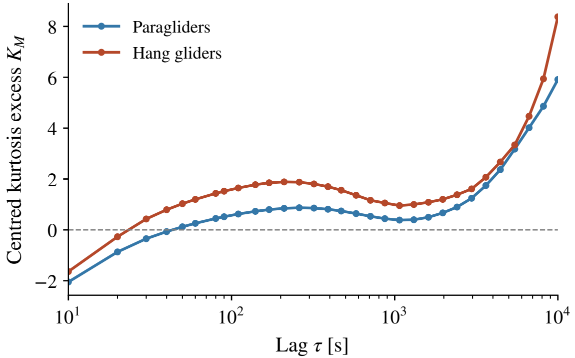
Can finite-record simulations resolve the predicted branch change under our sampling and selection?

Pooled windows. $\beta = 1.5$ is illustrative, not a fitted null; $q = \beta$ has a logarithmic correction. Zaburdaev, Denisov & Klafter (2015).

The moment spectrum challenges a specific Lévy-walk benchmark

The current pooled moment spectrum has $\zeta(2) = 1.737$ for paragliders and 1.682 for hang gliders. The corresponding $\zeta(q)/q$ ranges are imported in the report. The illustrative beta=1.5 line is not a fitted null. A finite-record comparison must reproduce sample lengths, cadence, selection and weighting, and assess whether the asymptotic branch change would be resolvable.

Non-Gaussian pooled increments can arise from Gaussian mixtures



$$\mathbf{Z} = \Sigma^{-1/2}(\mathbf{X} - \boldsymbol{\mu}),$$
$$K_M = \langle |\mathbf{Z}|^4 \rangle - 8.$$

A nonsingular 2D Gaussian has $K_M = 0$, including anisotropic Gaussians.

The observed curve changes sign and rises at large lag, where support is weakest.

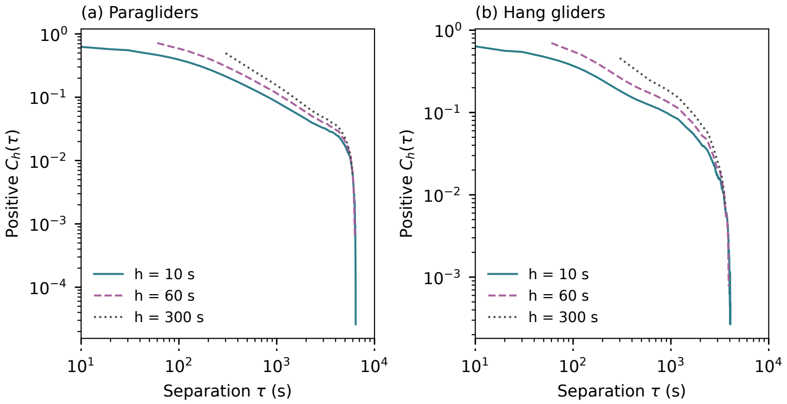
Test Gaussianity within matched environments or phases before rejecting conditional Gaussian dynamics.

Centred, full-covariance Mardia excess; pooled windows. Positive excess does not identify a power-law tail. Mardia (1970).

Non-Gaussian pooled increments can arise from Gaussian mixtures

Explain the full covariance normalization: unequal east and north variances alone do not produce a nonzero population excess for a Gaussian. But mixing Gaussian variances does. If X equals sqrt(A) times a fixed Gaussian vector, K equals eight times Var(A) divided by E(A) squared. This illustrates heterogeneity; it is not a correction formula for arbitrary flights. The finite-sample estimate need not vanish even for Gaussian data, and overlapping windows require a matched reference before significance claims.

Velocity memory remains visible after averaging over minutes

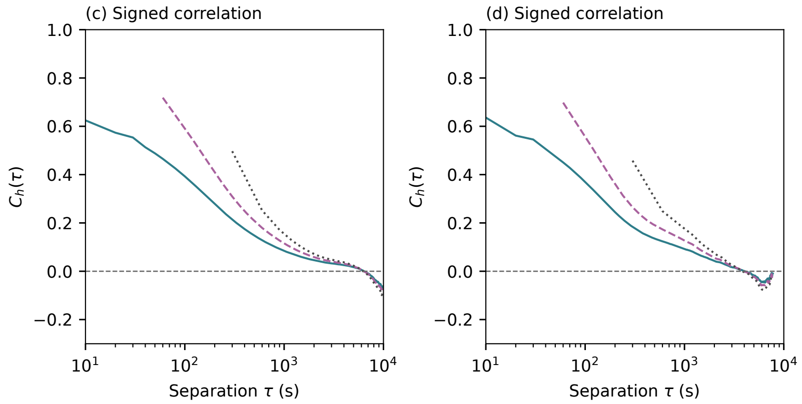


Do the curves support one decay law over the displayed scales?

Velocity memory remains visible after averaging over minutes

Use $\mathbf{v}_h(t) = [\mathbf{r}(t+h) - \mathbf{r}(t)]/h$ for h=10, 60, 300 seconds. Velocities use adjacent non-overlapping blocks; correlations are centred and normalized within flights before averaging. At 600 seconds the para values are 0.128, 0.180 and 0.252. These upper log panels show positive correlations only.

Signed correlations retain the long-lag negative values



What share of the long-lag decrease could follow finite-record centring or route returns?

Signed correlations retain the long-lag negative values

A log plot cannot show the negative part. Both disciplines have long-lag negative values in the signed representation. Coarse averaging can raise normalized correlations without increasing the underlying physical persistence time. At least twenty flights contribute to each plotted estimate.

Finite velocity persistence gives a ballistic-to-diffusive crossover

For the same stationary, centred velocity population,

$$M_{2,c}(\tau) = 2\sigma_v^2 \int_0^\tau (\tau - s) C(s)\, ds.$$

With $C(s) = e^{-s/t_p}$,

$$M_{2,c} = 2\sigma_v^2 t_p [\tau - t_p(1 - e^{-\tau/t_p})].$$

Ballistic at short times, diffusive at long times.

Taylor (1922), pp. 207–208. This covariance identity cannot be applied directly to differently weighted heterogeneous means.

Finite velocity persistence gives a ballistic-to-diffusive crossover

Which models are challenged, and which remain open?

Candidate	Prediction to test	Present conclusion
Homogeneous Brownian	Linear MSD; independent increments	Challenged over the measured range.
Constant drift + Brownian	Drift-free $V_2 \propto \tau$	Simple version challenged; mixtures unresolved.
Homogeneous fBm	Gaussian law; common H_p	Challenged as one pooled law.
Exponential velocity memory	Ballistic–diffusive crossover	Remains a candidate.
Renewal Lévy walk, $1 < \beta < 2$	Two-branch moment spectrum	Finite-record comparison still needed.

"Challenged" means a descriptive mismatch. Calibrated statistical rejection is still to be done.

Each candidate needs its own parameters, environmental assumptions and observation protocol. A long-time diffusive limit has not been tested.

Which models are challenged, and which remain open?

Ordinary Brownian motion is a poor descriptive model of the observed interval, but the asymptotic regime is unknown. fBm is challenged as a homogeneous pooled law; a phase-conditioned or environmentally heterogeneous Gaussian construction is not excluded. The exponential-persistence candidate deserves a joint MSD and VACF comparison. Standard Lévy walk needs finite-record calibration rather than rejection from an illustrative asymptotic line. An unbounded Lévy flight cannot literally describe continuous finite-speed flight; effective truncated versions need separate predictions.

Phase-conditioned moments must include windows crossing transitions

$$M_q(\tau) = \sum_c w_c(\tau) M_q(\tau \mid c), \qquad w_c = \frac{n_c}{\sum_{c'} n_{c'}}.$$

Here c partitions all admissible windows, including phase histories that cross boundaries. With equal flight weights, apply the identity within each flight before averaging.

The identity holds for any exhaustive labels. Recovering the global curve verifies accounting, not the physical accuracy of those labels.

Which episode durations and between-phase dependencies could distinguish renewal dynamics from persistent motion?

Phase-conditioned moments must include windows crossing transitions

Within-phase windows alone generally omit part of the global moment. Both the mixture weights and conditional moments depend on lag, so global exponents are not weighted averages of phase exponents. A glider moves while climbing; a climb is not automatically a waiting state.

Three decisions for this meeting

1. Define the model target.
   Ground-relative motion conditional on environment, or a residual after an independently justified wind correction?
2. Agree the first calibrated comparisons.
   Fit Brownian controls, fBm, finite persistence and a specified Lévy walk; simulate the same record lengths, selection and estimators.
3. Choose the most informative next measurements.
   Site–day uncertainty and route matching now; phase durations, increments and between-phase dependence after segmentation validation.

We have measured constraints on growth, shape, geometry and memory. The generating process and an intrinsic Hurst exponent remain unresolved.

Three decisions for this meeting

Agree the physical target, a small set of explicit candidate processes, and a sampling model for uncertainty. No rotation or rescaling has recovered still-air motion. Phase-conditioned comparisons should include transitions and between-episode dependence. Calibrated comparisons can begin before declaring a final physical model.

Squared laws transform the same distributional evidence

$$Q_{Y^2}(p) = Q_Y(p)^2, \qquad H_{Y^2,p} = 2H_{Y,p} \quad \text{for } Y \geq 0.$$

Squaring bin edges preserves their probability masses. A monotone transform also preserves the supremum distance between the correspondingly transformed CDFs.

If $f_Y(0^+) = c > 0$, then $f_{Y^2}(s) \sim c/(2\sqrt{s})$ near zero. This Jacobian effect says nothing about a heavy upper tail.

Which representation communicates the same result more clearly: displacement magnitude or its square?

Squared laws transform the same distributional evidence

The appendix includes the squared-law figures for both disciplines. Replotting the same observations after a monotone transformation is useful for connecting quantiles with second moments; it does not multiply the amount of independent evidence. Zero mass and dimensional units still need to be carried consistently.

Sources and the scope of the evidence

Zaburdaev, Denisov & Klafter (2015), *Lévy walks*	Moment-spectrum benchmark
Mandelbrot & Van Ness (1968), *Fractional Brownian motions*	fBm construction
Taylor (1922), *Diffusion by continuous movements*	Velocity/displacement relation
Mardia (1970), *Measures of multivariate skewness and kurtosis*	Centred kurtosis
Efron (1979), *Bootstrap methods*	Resampling principle

Numerical evidence: current Chapter 3 reports, frozen in the accompanying data and asset manifests. Bibliographic access limitations are retained in the repository audit; citations do not validate the chosen sampling design or constitute calibrated rejection tests.

Sources and the scope of the evidence

Joint laws and environmental interpretation: additional sources

Fraiman et al., Cramér–Wold results	Joint laws and projections
Didier & Pipiras, operator fractional Brownian motions	Beyond scalar rescaling
DHV classification guidance	Class is not a performance measure
DREAL Normandie, Le Pays de Caux	Coastal cliffs and valleys

The cited pages support specific mathematical or observational cautions. They do not identify the causes of the regional contrasts measured here.

Joint laws and environmental interpretation: additional sources